Large vs. Small LLMs as Evaluators of Abstractive Summarization

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Abstract. Large language models (LLMs) are increasingly used as automatic *judges* of text-generation quality because they align with human judgements far better than n-gram or embedding metrics. The strongest judges, however, are large and costly. We ask a practical question for abstractive summarization: how much evaluation quality is lost when a *small, open* LLM judge replaces a *large* one? Using a single LLM-as-a-judge pipeline, we score summaries from the SummEval benchmark on four dimensions—faithfulness, coherence, relevance and fluency—with a small open judge (Qwen2.5-3B, run locally) and a large proprietary judge (Google Gemini 2.5 Flash), each using an identical rubric and chain-of-thought prompt. We compare both against three baselines (ROUGE, BERTScore and a rubric-free zero-shot variant of the small model) on how well their scores correlate with human ratings, how closely the two judges agree, and their cost and speed. The large judge correlates with human ratings far better than every baseline (average Spearman 0.50 vs. at most 0.26), but the small judge does *not*: at 0.21 it is statistically indistinguishable from ROUGE and BERTScore and, strikingly, no better than its own rubric-free zero-shot variant—a rubric and chain-of-thought help the large model but not the small one. The small judge also shows a severe strictness bias, scoring up to 2.5 points lower than the large judge on faithfulness. We conclude that, on this benchmark, a small open judge is not yet a drop-in replacement for a large one; it offers only a cheap, rough signal. We report bootstrap confidence intervals throughout and release all code and prompts for reproducibility.

Keywords: LLM-as-a-judge · summarization evaluation · SummEval · chain-of-thought · small language models.

1 Introduction

Automatic evaluation of generated text is a long-standing bottleneck in natural language generation. For abstractive summarization, lexical-overlap metrics such as ROUGE [4] and embedding-based metrics such as BERTScore [9] are cheap and reproducible, but they correlate only weakly with human judgements of qualities such as factual consistency or coherence [1]. The "LLM-as-a-judge" paradigm—prompting a strong language model to rate or compare outputs—has

emerged as a much stronger alternative, matching human preferences closely on a range of tasks [10, 5, 2]. The catch is that the best judges are large, often proprietary, models that are expensive, rate-limited and difficult to reproduce.

This motivates a simple, practical question:

> *Does replacing a large LLM judge with a small, open one meaningfully hurt evaluation quality, and by how much?*

This is an *evaluation* study, not a generation study: the system never writes summaries. It scores summaries that already exist in the SummEval benchmark [1] and measures how well those scores track human ratings. We instantiate one LLM-as-a-judge pipeline and run it with two judges of very different size and access model—a small open judge, Qwen2.5-3B [7] served locally, and a large proprietary judge, Google Gemini 2.5 Flash [3], accessed through its API—under an identical rubric and chain-of-thought (CoT) prompt [8]. Both are compared with three baselines: ROUGE, BERTScore, and a rubric-free zero-shot scoring variant of the small model that isolates the contribution of prompt design.

Contributions. (1) A controlled comparison of a small open vs. a large proprietary LLM judge for summarization, holding the prompt, dataset and parsing fixed. (2) A three-way analysis—human alignment, judge–judge agreement, and cost/speed—with bootstrap confidence intervals to account for the modest subset size. (3) A fully open, reproducible pipeline (local small judge via Ollama; committed prompts, seeds and code).

2 Related Work

Reference-based metrics. ROUGE [4] measures n-gram overlap with reference summaries and remains a default in summarization. BERTScore [9] replaces surface overlap with similarity between contextual BERT embeddings, improving correlation with human judgement. Both are *reference-based*: they compare a candidate to gold summaries rather than judging it against the source, which limits their sensitivity to faithfulness.

Meta-evaluation of summarization. SummEval [1] re-evaluated summarization metrics on outputs of modern systems for CNN/DailyMail articles, collecting expert annotations along coherence, consistency, fluency and relevance. It showed that many automatic metrics correlate weakly with human scores and provides the human ratings we use as ground truth.

LLMs as judges. G-Eval [5] uses GPT-4 [6] with chain-of-thought and a form-filling protocol, reaching far higher human correlation than prior metrics. GPTScore [2] frames evaluation as the model's likelihood of a quality-conditioned continuation. Zheng et al. [10] study LLM judges for open-ended chat, documenting strong agreement with humans alongside biases (position, verbosity, self-enhancement).

Most of this work relies on large proprietary models such as GPT-4 [6] or Gemini [3]. We focus instead on whether a *small open* model can stand in for a large one, which determines whether LLM-based evaluation is affordable and reproducible at scale.

3 Methodology

3.1 Task Formulation

Given a source document d, a machine summary s and a scoring rubric, a judge produces four integer scores in $\{1, \ldots, 5\}$ for *faithfulness*, *coherence*, *relevance* and *fluency*. We treat the judge's faithfulness score as the counterpart of SummEval's *consistency* dimension (factual consistency with the source); the other three names align directly. We do not generate summaries; we only score existing ones and correlate the scores with human ratings.

3.2 Judges

Both judges receive an identical prompt and decode at temperature 0 for deterministic scores. The **small judge** is Qwen2.5-3B-Instruct [7], an open model run locally through an Ollama server (4-bit quantization) on a single consumer GPU—representing the cheap, reproducible regime. The **large judge** is Google Gemini 2.5 Flash [3], a large proprietary model accessed through its API, matching the proposal's intent of contrasting a small local judge with a large proprietary one. The variable of interest is the judge's *scale and capability*.

3.3 Rubric and Chain-of-Thought Prompt

A single prompt template defines the four dimensions, instructs the model to reason briefly (2–4 sentences) before scoring, and to emit a final JSON object with the four integer scores. Chain-of-thought reasoning [8] before the numeric verdict is the key prompt-design element shared by both judges; the prompt is identical across judges and committed verbatim in the repository.

3.4 Baselines

ROUGE [4] (ROUGE-1/2/L F-measure, max over references) and **BERTScore** [9] (F, RoBERTa-large) are reference-based. **Zero-shot Direct Scoring** uses the *same* small model prompted for a single overall 1–5 score with no rubric and no reasoning; comparing it with the rubric+CoT small judge isolates how much of the small judge's quality comes from prompt design rather than raw model ability.

3.5 Output Parsing

Small models occasionally emit malformed output. The judge output is parsed by first extracting the trailing JSON object and, failing that, a per-dimension regular expression; up to two re-queries are issued before a record is marked unparsed. This structured-output-plus-fallback strategy keeps parse failures negligible.

4 Experimental Setup

4.1 Dataset

We use SummEval [1] (the `mteb/summeval` release, 100 articles, each with 16 machine summaries and expert human ratings on the four dimensions). We build a fixed, seeded evaluation subset of all 100 articles, sampling 8 of the 16 machine summaries per article, yielding 800 summary-level records. The subset is regenerated deterministically from a fixed seed.

4.2 Metrics

Human alignment is measured by Spearman's ρ and Kendall's τ between each method's scores and the human ratings, per dimension and averaged. For the two LLM judges each dimension is matched to its human counterpart; the single-value methods (ROUGE, BERTScore, zero-shot) are correlated against each human dimension, as is standard for reference metrics. Judge–judge agreement is measured by Pearson correlation and the mean (signed and absolute) score difference. Because the subset is modest, we report **95% bootstrap confidence intervals** (1000 resamples over records) for the average correlations. Cost is reported as tokens per summary, latency and throughput.

4.3 Implementation Details

The small judge and zero-shot baseline run on Qwen2.5-3B via Ollama; the large judge runs on Gemini 2.5 Flash via the Google Gemini API with client-side rate limiting. BERTScore runs on CPU to avoid contending with the local judge for GPU memory. Long source bodies are lightly truncated to fit the small model's context window. All decoding uses temperature 0; seeds, prompts and pinned dependencies are committed. Each stage writes its scores to disk so later stages and analyses are fully reproducible from the released artifacts.

5 Results

5.1 Human Alignment

Table 1 reports per-dimension and average Spearman/Kendall correlations with human ratings, Table 2 the average Spearman with 95% bootstrap confidence

Table 1. Spearman correlation of each method with SummEval human ratings, per dimension. Avg. columns report the mean Spearman (ρ) and Kendall (τ) across the four dimensions. LLM judges in bold.

Method	Coh.	Cons.	Flu.	Rel.	Avg. ρ	Avg. τ
ROUGE-1	0.214	0.186	0.102	0.313	0.204	0.151
ROUGE-2	0.144	0.150	0.076	0.221	0.148	0.109
ROUGE-L	0.212	0.153	0.078	0.230	0.168	0.124
BERTScore-F	0.385	0.075	0.092	0.373	0.231	0.169
Zero-shot	0.296	0.238	0.232	0.278	0.261	0.226
Small judge	0.244	0.261	0.128	0.210	0.210	0.183
Large judge	0.506	0.621	0.417	0.452	0.499	0.434

Table 2. Average human Spearman correlation with 95% bootstrap confidence intervals (1000 resamples over the evaluation records).

Method	Avg. Spearman	95% CI
ROUGE-1	0.204	[0.156, 0.251]
ROUGE-2	0.148	[0.094, 0.195]
ROUGE-L	0.168	[0.120, 0.216]
BERTScore-F	0.231	[0.185, 0.281]
Zero-shot	0.261	[0.212, 0.311]
Small judge	0.210	[0.171, 0.250]
Large judge	0.499	[0.459, 0.536]

intervals, and Fig. 1 the average alignment. The **large judge** is the clear winner: it has the highest correlation on every dimension and an average Spearman of 0.50—roughly double the best baseline—with a confidence interval ($[0.46, 0.54]$) that does not overlap any other method. It is strongest on consistency/faithfulness ($\rho = 0.62$), precisely the dimension that reference metrics capture worst.

The **small judge** tells a very different story. Its average correlation (0.21) is statistically indistinguishable from ROUGE-1 (0.20) and BERTScore (0.23)—the confidence intervals overlap almost entirely—so rubric-guided judging by a 3B model buys essentially nothing over cheap reference metrics here. More strikingly, the small *rubric+CoT* judge (0.21) is no better than, and in point estimate slightly below, the *rubric-free* zero-shot variant of the same model (0.26). The elaborate rubric and chain-of-thought that help the large model therefore do *not* help the small one: a 3B model scores summaries about as well with a bare "rate 1–5" prompt. On this benchmark, human alignment is driven far more by judge *scale* than by prompt design.

5.2 Judge–Judge Agreement

Table 3 reports agreement between the two judges. Their scores correlate only weakly (overall Pearson 0.22), and the small judge is systematically and substantially *stricter*: it scores about 1.5 points lower than the large judge on average,

Table 3. Agreement between the small and large judge per dimension: Pearson correlation, mean signed difference (small−large; negative means the small judge is stricter) and mean absolute difference.

Dimension	Pearson	Mean diff.	Mean abs. diff.
Faithfulness	0.337	-2.473	2.527
Coherence	0.194	-1.653	1.767
Relevance	0.232	-0.762	1.046
Fluency	0.208	-1.294	1.413
Overall	0.221	-1.546	1.688

Table 4. Cost and speed of the model-based methods on the evaluation subset.

Method	N	Tok./summ.	Lat. (s)	Summ./min	Monetary
Zero-shot (local)	800	615	0.85	70.1	\$0 (local GPU)
Small judge (local)	800	1122	6.20	9.7	\$0 (local GPU)
Large judge (Gemini)	800	931	1.10	54.6	∼\$0.40

rising to 2.5 points on faithfulness, where it assigns low scores even to summaries the large judge and human annotators consider consistent. This large, systematic strictness/compression bias—not random noise—is the small judge's dominant failure mode. Because it is systematic, simple score *calibration* (e.g. per-judge standardisation) is a plausible partial remedy, though it cannot recover rank information the small judge never produced.

5.3 Cost and Speed

Table 4 summarises the cost–quality trade-off, and the two regimes differ sharply. The small judge runs entirely locally at zero API cost but is slow on a consumer GPU (∼6 s per summary, 9.7 summaries/min). The large judge, on a paid but inexpensive API, is both faster (∼1 s per summary) and far more accurate, costing only about US\$0.40 for all 800 summaries (∼US\$0.0005 each). On this benchmark the large judge therefore dominates on quality at a near-trivial price; the small judge's appeal is confined to settings where local execution or zero marginal cost outweighs a large loss in evaluation quality.

6 Discussion

Our results give a clear, and largely negative, answer to the motivating question. On SummEval, replacing a large judge with a small open one does *not* merely cost a little quality—it erases most of the benefit of LLM-as-a-judge: the small judge's human alignment (0.21) falls back to the level of the ROUGE/BERTScore baselines it was meant to surpass, while the large judge (0.50) stands well above everything. Two findings sharpen this picture. First, rubric and chain-of-thought

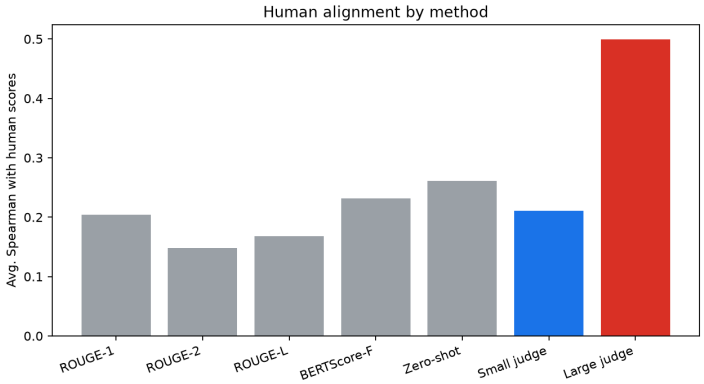


Fig. 1. Average Spearman correlation with human ratings by method. The large judge (red) dominates every method, while the small judge (blue) is only on par with the reference-based and zero-shot baselines.

prompting—central to methods such as G-Eval [5]—are *not* model-agnostic: they help the large judge but give the 3B model no advantage over a bare direct score, suggesting the small model lacks the capacity to exploit the structured protocol. Second, the small judge's errors are dominated by a large, systematic strictness bias rather than random noise, which calibration can dampen but not remove. For practitioners, the implication is that LLM-as-a-judge for summarization currently hinges on judge scale: a small local judge gives a cheap, rough signal, but trustworthy automatic evaluation still needs a large judge—which, encouragingly, costs only cents per several hundred summaries.

7 Limitations

The evaluation subset, though seeded and spanning all SummEval articles, is modest; we therefore report bootstrap confidence intervals, which are non-trivially wide and should temper strong rankings. SummEval covers a single domain (CNN/DailyMail news). Our large judge is Gemini 2.5 Flash, a capable but not frontier-scale model, and the two judges differ in architecture and training as well as size, so the comparison reflects a realistic small-open vs. large-proprietary contrast rather than a pure parameter-count ablation; results may differ for the very largest judges such as GPT-4 [6]. Light source truncation may remove information relevant to faithfulness. Finally, scores at temperature 0 are deterministic but do not capture sampling variance.

8 Conclusion

We compared a small open judge (Qwen2.5-3B, local) and a large proprietary judge (Gemini 2.5 Flash) for abstractive-summarization evaluation under an

identical rubric and chain-of-thought prompt, against ROUGE, BERTScore and a rubric-free baseline, on a seeded SummEval subset of 800 summary-level records. The large judge correlates with human ratings far better than every other method (average Spearman 0.50), but the small judge does not: at 0.21 it is indistinguishable from the classical metrics and no better than its own rubric-free variant, and it suffers a severe strictness bias. The quality gap thus tracks judge scale, and prompt sophistication does not close it. We conclude that a small open judge is not yet a drop-in replacement for a large one in summarization evaluation, even though the large judge's monetary cost is negligible. All code, prompts and configuration are released for reproducibility.

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